

DNA Vaccines

A Review of Developments

Robert G. Webster¹ and Harriet L. Robinson²

- 1 St. Jude Children's Research Hospital, Department of Virology and Molecular Biology, Memphis, Tennessee, USA
- 2 University of Massachusetts Medical Center, Department of Pathology, Worcester, Massachusetts, USA

Contents

Summary	273
1. Historical	275
2. Routes and Methods of Immunisation	277
3. Uptake and Expression	279
4. Plasmids for DNA Immunisation	280
5. Immunity to Infectious Agents Induced by DNA Immunisation	282
6. Advantages and General Features of DNA Immunisation	282
6.1 Induction of Protective Immunity to Influenza in Mice and Chickens	282
6.2 Induction of Crossreactive (Heterotypic) Immunity to Influenza Viruses	282
6.3 Induction of Long Term B Cell Immunity to Influenza in Mice	283
6.4 Broadening of the Antibody Response to the Haemagglutinin of Influenza Virus	283
6.5 Modulation of the Antibody Isotype by DNA Immunisation	283
6.6 Manipulation of the Immune Response with Cytokine Genes	284
6.7 DNA Immunisation Can Overcome Haplotype Restriction	284
7. DNA Immunisation to HIV	285
8. Variability in the Immune Responses Induced by DNA Immunisation	285
9. Safety Considerations	286
9.1 Anti-DNA Antibodies	286
9.2 Altered Immune State	287
9.3 Transformation	287
9.4 Regulatory Considerations	288
10. Mechanisms of Induction of Immunity by DNA Immunisation	288

Summary

Immunisation with purified DNA is a powerful technique for inducing immune responses. The concept is very simple, involving insertion of the gene encoding the antigen of choice into a bacterial plasmid, and injection of the plasmid into the host where the antigen is expressed and induces humoral and cellular immunity. This technology can induce immunity to all antigens that can be encoded by DNA; this includes all protein, but not carbohydrate, antigens. DNA immunisation appears to result in presentation of antigens to the host's immune system in a natural form, similar to that achieved with live attenuated vaccines.

The most efficacious routes for DNA immunisation are bombardment with particles coated with DNA (gene-gun), followed by intramuscular and intradermal

administration. The efficiency of transfection of host cells is low, but sufficient to induce immunological responsiveness. The DNA plasmid is retained in the transfected cells in an unintegrated form for the life of the cell. The majority of transfected cells are eliminated, but residual expression has been detected for longer periods.

In animal model systems, DNA immunisation has been shown to induce protective immunity to influenza, herpes, rabies, hepatitis B and lymphocytic choriomeningitis viruses, and to malaria and mycobacteria. However, strategies to induce protective immunity to HIV and other disease agents remain to be developed. DNA vaccines permit modulation of the immune response by altering the route or method of DNA administration, by including immunostimulatory sequences in the plasmid, and by co-administration of cytokine genes with the gene encoding the antigen of interest. A T helper 1 response provides cell-mediated immune killing of infected cells and neutralising antibody production, while a T helper 2 response induces IgE and allergic responses.

The advantages of DNA immunisation are: (i) similarity to live attenuated vaccination but without the possibility of contamination with undesirable agents; (ii) correct presentation of antigen; (iii) combinations of DNA-encoded antigens and/or cytokines may be administered; (iv) genetic stability; (v) potential speed of making new vaccines with genetic identity; (vi) development of vaccines for agents that cannot be grown in culture; (vii) no need for a cold chain; and (viii) possibility of modulation of the immune response. The perceived risks include: (i) integration of the plasmid into the host genome; (ii) induction of anti-DNA antibodies and autoimmunity; and (iii) induction of tolerance. The available information concerning safety is encouraging, with the risk of integration being considered to be orders of magnitude below the spontaneous mutation frequency in humans.

DNA immunisation offers the possibility of extending the control of infectious diseases far beyond those that are currently controlled by conventional and recombinant vaccines, to include vaccines for parasites and cancer. However, it is currently too early to predict the future extent of use of DNA vaccines in human immunisation programmes because the initial clinical trials are still in progress.

The eradication of smallpox, the control and likely eradication of poliomyelitis and the control of measles have been achieved by vaccination, and the cost effectiveness amounts to billions of dollars per year. The quest to control a large number of infectious diseases continues, for the fact is that vaccines are not available for most infectious diseases and conventional vaccine strategies are unlikely to produce such vaccines. Worldwide, tuberculosis kills at least 3 million people per year and the incidence is increasing. Infectious intestinal diseases cost approximately \$US28 billion in med-

ical costs and lost productivity in the US per year. Approximately 300 million people are infected with malaria worldwide, and the direct cost of treating influenza in the US is \$US4.6 billion per year.^[1] Currently, only about 25 major human diseases are controlled by vaccines, leaving a large number of bacterial, fungal, parasitic and viral diseases uncontrolled. In addition, the immune modulation of tumours is largely unresolved. The inability to cultivate organisms such as syphilis, hepatitis C, malaria and leprosy in sufficient quantity has greatly hampered vaccine development

against these diseases. Alternative strategies are therefore needed that can meet the needs of an expanding world population (see fig. 1).

A novel vaccine strategy that has enormous potential is the use of pure DNA for immunisation. This article will consider this new method of immunisation and its potential as a future vaccine strategy that would be applicable to diseases for which vaccines are currently not available and to

improving current vaccines, as well as to resolving fundamental unanswered questions concerning the induction and maintenance of immunity. Knowledge in this burgeoning field is also applicable to the parallel field of gene therapy.

1. Historical

Cells were first inoculated with isolated genetic information in 1956, when Gierer and Schramm^[2]

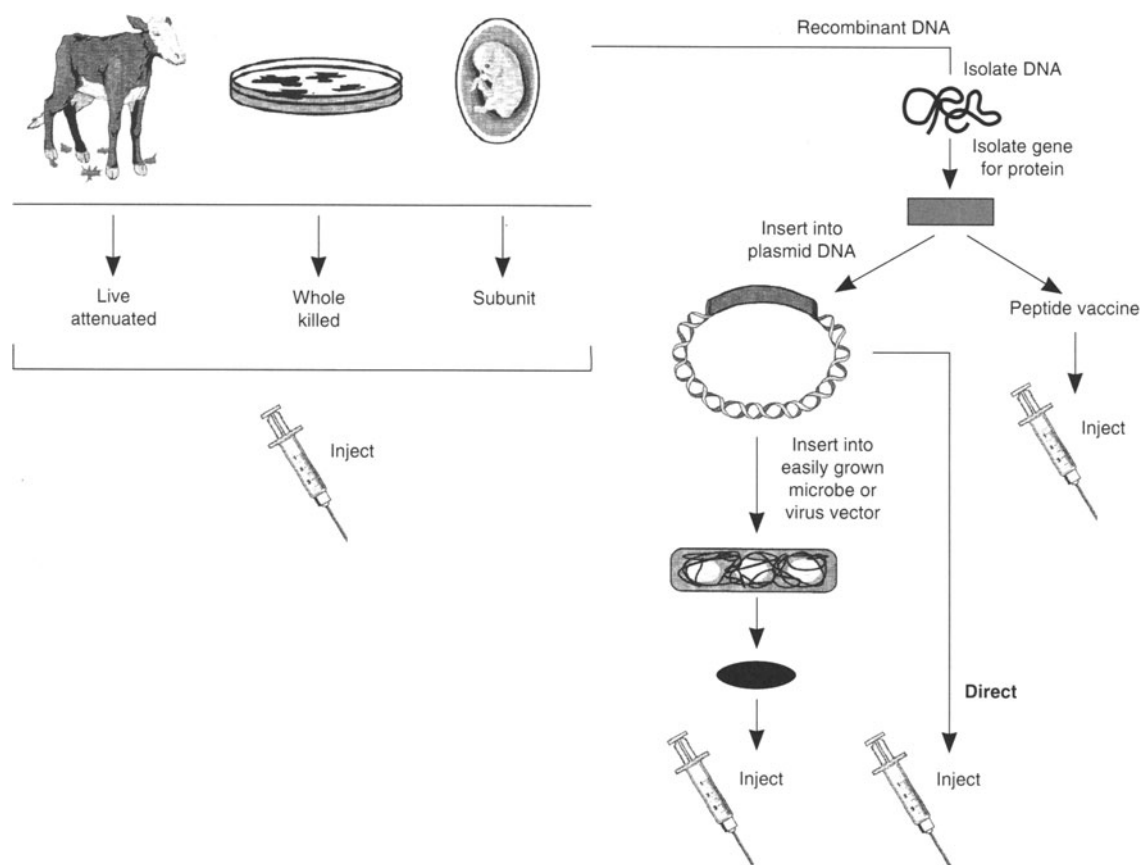


Fig. 1. Current and future vaccine strategies. **Live attenuated** agents originate from another host or are derived by passage in an alternative host (e.g. smallpox was first controlled by vaccination with live cowpox virus). **Whole killed** vaccines are prepared from the inactivated disease agent (e.g. poliomyelitis virus grown in cell culture and inactivated with formaldehyde). **Subunit** vaccines contain proteins isolated from disrupted organisms (e.g. purified influenza virus is disrupted with detergent or ether and the surface glycoproteins used for immunisation). More recently, the genes encoding the antigens have been isolated. Peptides comprising a portion of or the entire epitope have been synthesised and used as experimental vaccines (e.g. the linear epitope on foot and mouth disease virus). **Recombinant** vaccine strategies are used to insert the gene coding for an antigen into a suitable bacterium for expression, and the purified antigen is used as vaccine. Alternatively, the sequences are incorporated into bacteria or viruses (e.g. poxviruses or *Shigella*) that are used to infect the host, where they express the desired antigen. DNA vaccination involves the direct injection of a DNA plasmid encoding the antigen of choice.

demonstrated that isolated RNA from tobacco mosaic virus (TMV) induced lesions in tobacco plants. The efficiency of infection with TMV RNA was low (1.8%) but clearly demonstrable. Subsequently, RNA from poliomyelitis virus was demonstrated to be infectious in cultured animal cells,^[3] and in 1981 Racaniello and Baltimore^[4] showed that a complementary DNA for poliovirus was infectious in mammalian cells. Later, it was shown that direct introduction of genes into chickens or rats resulted in protein expression.^[5,6] Studies with isolated DNA from polyoma virus demonstrated the induction of tumours in newborn hamsters and the induction of antibodies.^[7,8]

The use of direct DNA inoculation for the induction of immune responses was initiated quite independently by 3 research groups in the early 1990s: (i) Wolff/Vical/Merck; (ii) Johnston and Sanford; and (iii) Robinson and Webster. Felgner and Malone of Vical, working with Wolff's group from Wisconsin, were working to increase DNA uptake into muscle cells by using liposomes. Unexpectedly, the control DNA, containing no liposomes and hence termed 'naked' DNA, produced higher levels of protein than the formulated DNA.^[9] Several groups, including Felgner and Malone, realised the importance of this finding for DNA-based immunisation. Early successes were realised at Merck by Liu and co-workers,^[10] at the Saskatchewan Agriculture Station by Babiuk and co-workers,^[11] at Pasteur by Whalen and Davis^[12] and at the University of Pennsylvania by Weiner and co-workers.^[13] Johnston, a co-inventor of the gene-gun developed to shoot DNA into plant cells, had been using the gun to deliver genes for gene therapy and genetic immunisation. However, the recipient mice had not been tested for antibodies because the amount of administered growth hormone (nanogram quantities) was considered below the level considered sufficient to induce an immune response. With the screening of the mice for antibody, it was discovered that the mice were seropositive, and that very low amounts of a DNA-expressed antigen could raise an antibody response.^[14]

In 1988, we reported that a retrovirus-expressed haemagglutinin of influenza virus could protect against lethal influenza virus infection.^[15] Protection was achieved by immunisation with a replication-competent haemagglutinin-expressing retrovirus. Surprisingly, protection was also achieved by the inoculation of a replication-defective haemagglutinin-expressing retrovirus that had been recovered as an infectious agent using a packaging cell line (unpublished observations). Remarkably few infectious units of the packaged-defective virus ($<1 \times 10^6$) were required to elicit protective immune responses. In view of this, it was decided to test whether the plasmid DNA that encoded the defective H7-expressing virus could be used for vaccination. In this case, successful immunisation would require the uptake and expression of the inoculated plasmid DNA. In the very first experiment, 6 of 6 chickens were protected from lethal challenge with a fowl plague H7N7 influenza virus, while control chickens receiving the plasmid without the H7 insert all died. As is often the case with novel findings, the scientific community was highly sceptical of our results and the journal *Science* rejected the initial report from Robinson, Hunt and Webster in March 1992. This report subsequently appeared in *Vaccine*,^[16] and was the first accepted paper to demonstrate the induction of protective immunity by DNA immunisation.

At the same time, Liu and colleagues at Merck followed up on the studies from Vical, using a plasmid containing a DNA copy of the nucleoprotein of influenza virus to inoculate mice intramuscularly. The mice produced antibodies and cytotoxic T lymphocytes (CTLs) to the nucleoprotein, and 90% of the mice were protected against challenge with different subtypes of influenza virus.^[10] Subsequent studies by Wang et al.^[13] reported the induction of both antibodies and CTLs against the gp120 and Rev proteins of HIV-1 after injection of mice with plasmid DNA encoding these proteins. These reports were followed by the demonstration of induction of immune responses to hepatitis B surface antigen^[12] and bovine herpes virus.^[11] By 1994, there was guarded recognition of the concept

2. Routes and Methods of Immunisation

The routes of inoculation used for DNA immunisation include intramuscular, intradermal and gene-gun delivery to the skin.^[9,10,14,16] The initial studies showed that injection of mouse skeletal muscle with RNA or DNA expression vectors containing the genes for chloramphenicol acetyltransferase, luciferase and β -galactosidase resulted in protein expression.^[9] Subsequent studies by others^[17] demonstrated that injections by other routes were ineffective, and that only striated muscle was capable of taking up and expressing reporter genes.^[12] In contrast, studies on induction of immunity by injection of DNA has revealed that a number of routes were efficacious at inducing immunity. Thus, intravenous, intranasal, intradermal and subcutaneous routes as well as intramuscular routes could be used for administering DNA encoding the haemagglutinin of influenza virus and for induction of protective immunity in mice.^[18] Use of a gene-gun to deliver DNA-coated gold beads into the skin of mice proved to be the most efficient method of raising protection, requiring 100- to 1000-fold lower amounts of DNA than injections of DNA in saline. Thus, different routes as well as methods of inoculation of DNA could induce immunity, suggesting that features other than long-lived expression of the injected DNA in muscle were important for the induction of immune response.

Although different routes and methods of inoculation of DNA can result in the induction of immunity, a key issue is the type of cells that are transfected and how professional antigen-presenting cells (APCs) acquire the antigen for presentation to T cells. To date, most work has been done on antigen presentation of the DNA inoculated in saline into the muscle, or by gene-gun delivery of DNA precipitated on gold beads into the skin. After DNA injection in saline into the muscle, the DNA is taken up from the intracellular spaces and approximately 1.5% of muscle cells express the gene product.^[9] The actual mechanism of transfection of cells from the intracellular species is still poorly understood. Myocytes, as far as is known, do not

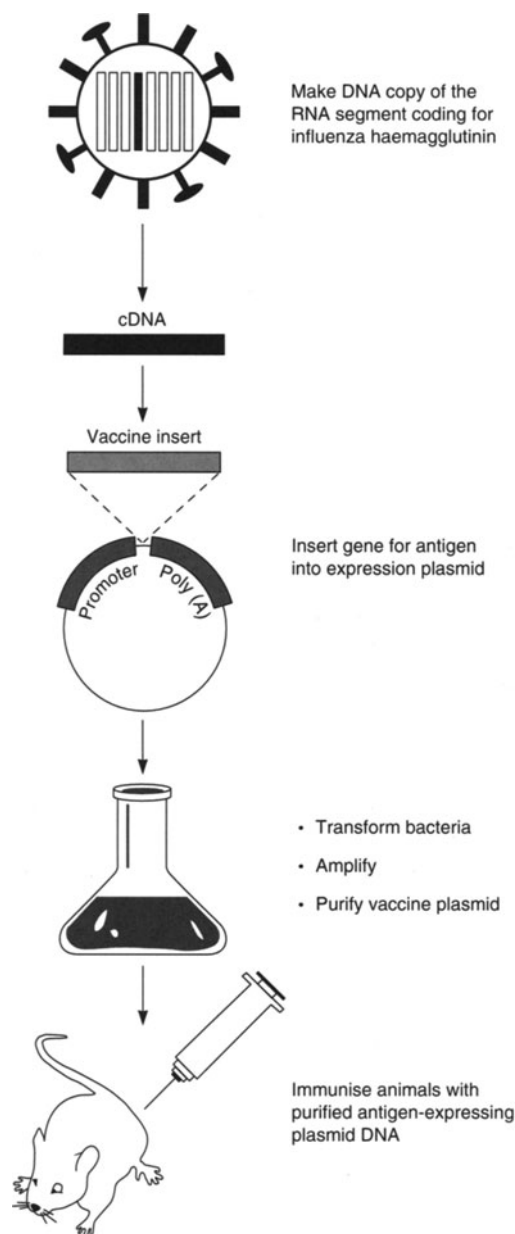


Fig. 2. Preparation of a DNA vaccine to influenza virus haemagglutinin.

of DNA immunisation as a new strategy for induction of protective immune responses against infectious agents and a new method for studying the *in vivo* function of the immune system (table I).

Table 1. Early examples of induction of immunity to infectious agents by DNA immunisation. This table is not designed to be all-inclusive

Disease agent	Gene	Animal	Antibodies	Cell-mediated response	Protection	Comment	Reference
Influenza (H7N7, H1N1)	Haemagglutinin, nucleoprotein	Chicken, mouse	Yes	CTLs to nucleoprotein	Protection from challenge	Homologous and heterologous protection	10, 16, 18
Herpes bovine	BHV-1 glycoprotein	Mice, cattle	ELISA antibodies	NT	Reduced virus shedding in calves	Seroconversion in both mice and cattle	11
Hepatitis B	HBsAg	Mice	High levels of circulating antibodies	Strong CTL responses	NT	Superior to recombinant protein immunisation	12
Rabies	Rabies glycoprotein	Mice	Neutralising antibodies	Virus-specific CTLs	Protection from lethal challenge	Full response involving cytotoxic and helper T cells	58
HIV	<i>env</i> , <i>gag</i> , <i>pol</i> , <i>rev</i>	Mice, monkeys	Antibodies to gp160 and gp41	Lymphocyte proliferation and CTLs	NT	Cross clade antibody activity, no challenge	13, 108
LCMV	Nucleoprotein, glycoprotein	Mice	Antiviral antibodies	Virus-specific CTLs	Reduced virus titres	Less efficacious than recombinant vaccinia expressed antigens	60, 61, 62
Herpes	Glycoprotein B	Mice	HSV-specific antibody of IgG2a isotype	Adoptive transfer with CD4+ cells; no CD8+ response	Protection against persistent infection	Immunity similar to infectious virus	57
<i>Plasmodium yoelii</i> (malaria)	Circumsporozoite protein (PcSP)	Mice	High levels of antibodies	Strong CTL responses	Yes	9 of 16 mice protected superior to irradiated sporozoite immunisation	109
<i>Mycobacterium leprae</i>	<i>M. leprae</i> heat shock (MLhsp65)	Mice	NT	CD4+ and CD8+ T cell responses	Yes	Adoptive transfer of immunity with CD8 cells	110

Abbreviations: CTL = cytotoxic T lymphocyte; ELISA = enzyme-linked immunosorbent assay; HBsAg = hepatitis B virus surface antigen; HSV = herpes simplex virus; LCMV = lymphocytic choriomeningitis virus; NT = not tested.

express the major histocompatibility complex (MHC) class II molecules that are important for presenting antigen to CD4+ T cells, nor do they express the costimulating molecules, such as B7-1/B7-2, that are necessary for optimal priming of CD8+ CTLs.^[19-21] Professional APCs, such as macrophages or tissue dendritic cells, in the muscle may be directly transfected.^[22] Following intradermal injection in saline or gene-gun delivery, the DNA is expressed primarily in keratinocytes.^[23-25] After gene-gun delivery to the skin, dendritic cells containing antigen have been detected in the draining lymph nodes.^[26] Gene-gun administration of DNA can result in direct transfection of cells.^[12,23] Expression is markedly reduced after 7 to 10 days, due to the normal sloughing of skin. The epidermis appears to play a major role in gene-gun DNA immunisation, since DNA-coated gold beads deposited primarily in the dermal layer of the skin produce a lower antibody response.^[23] It has been suggested that Langerhans cells within the epidermis, or even dermal macrophages, may act as APCs for intradermal and gene-gun DNA immunisation.^[18,22,24]

The efficiency of gene expression is remarkably low, even after intramuscular injection, with as few as 100 muscle fibres showing expression along the needle track of injection.^[27] Consequently, attention has focused on methods to enhance expression in target tissues. In muscle cells, expression is reportedly higher in regenerating cells, and expression has been enhanced by preinjection of hypertonic sucrose,^[12] bupivacaine^[28] or snake venom cardiotoxin^[29] that damage muscle at the site of injection, leading to regeneration and the presence of immature susceptible cells.

One of the most effective methods for potentiating the efficiency of transfection is the use of a gene-gun to shoot DNA-coated gold beads directly into cells.^[30] This method has been shown to be effective in a number of somatic tissues both *in vivo* and *in vitro*. Expression of the firefly luciferase gene was detected in 10 to 20% of liver and skin cells at the site of bombardment; the microprojectiles penetrated multiple cell layers but pro-

duced no evidence of tissue damage.^[31] In the dermis of rats, sustained luciferase activity was demonstrated for up to 1.5 years after gene-gun delivery.^[32] The most dramatic effect of particle bombardment with DNA was found on induction of immunity after DNA transfer to the epidermis.^[18] As little as 0.04 µg of DNA encoding the haemagglutinin of influenza A/PR/8/34 (H1N1) provided 100% protection from challenge after 1 immunisation. Studies with influenza nucleoprotein administered to the epidermis of mice demonstrated the induction of both humoral immunity and CTLs following immunisation of mice with as little as 16 ng of DNA in a single dose.^[33] In contrast, 5000-fold more DNA was required to induce comparable results by intramuscular and intradermal injections of DNA.

The use of attenuated *Shigella* provides a novel method for delivering plasmid DNA into host cells.^[34] Highly attenuated *Shigella*, a bacterial vector that naturally enters epithelial cells and escapes the phagocytic pathway, delivers plasmid DNA to the cytoplasm of the host cell. The advantages of this system are that the *Shigella* vector is designed to deliver DNA to the colonic mucosa and

is, therefore, suitable for oral or mucosal immunisation. Sizemore et al.^[34] propose that delivery of DNA-encoded antigens to the mucosal immune system should permit mucosal immunisation simultaneously with multiple antigens that: (i) can be directed for class I presentation, class II presentation, or both; (ii) can stimulate T helper cells types 1 or 2 (T_H1 or T_H2); and (iii) can be secreted while maintaining the proper folding and conformational epitopes for IgA and IgG antibody production. Additionally, since no DNA purification is required for this type of DNA vaccination, the costs involved are minimal. This method of delivering DNA has special applicability to bacterial and viral diarrhoeal diseases, and establishes the principle of delivering DNA plasmids to particular host cells depending on the vector system employed.

Other methods to potentiate DNA uptake and expression include jet injection with 'needle free' injection devices,^[35] cationic lipids for delivery to the lungs, other liposome systems, and ligand-mediated gene delivery.^[27] Each of these modalities have been tested with some success for gene therapy, but have not been sufficiently evaluated for potentiation of DNA immunisation. It is apparent that both the route and method of injection are important considerations for DNA immunisation, and that the level of expression, while important, is less critical than making optimal contact with APCs (see section 10). Another consideration for optimising DNA vaccines is through the adjuvant effects provided by immunostimulatory sequences in the plasmid construct (see section 4).

3. Uptake and Expression

The mechanism of DNA uptake by cells has not been resolved. In muscle cells, it has been proposed that DNA is taken up through the T-tubules.^[36] In other cells, including the thyroid and synovium, it is proposed that DNA is taken up by normal endocytosis^[37,38] into coated or uncoated vesicles. DNA, when administered by the particle bombardment method, is delivered directly into the cells. Overall, the efficiency of direct DNA

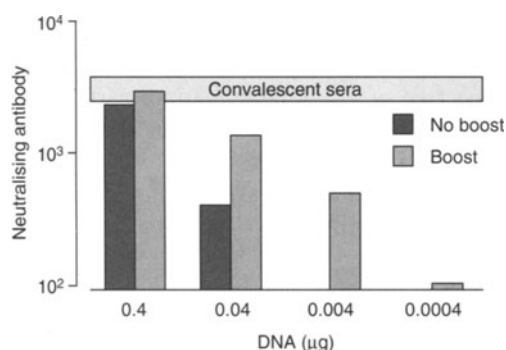


Fig. 3. Induction of neutralising antibody following a single gene-gun inoculation. Values are geometric mean titres 6 months after injection of influenza H1 DNA vaccine. Convalescent sera represent the range of titres observed in mice at 2 weeks post-intranasal challenge with 400 plaque-forming units (pfu) of A/PR/8/34 (H1N1). The neutralisation assay scored for complete neutralisation of 100 pfu of A/PR/8/34 in a microtitre assay on MDCK cells.^[89]

administration is very low, and until methods are developed for increasing the efficiency, direct gene inoculations will have more potential for inducing immune responses than for gene therapy. For immunological priming and maintenance, a small number of cells transiently expressing antigen may be ideal; however, higher levels of sustained expression are required for gene therapy.

It might be anticipated that plasmid DNA that enters a host cell would be integrated into the cell chromosome at low frequency.^[39] However, the available evidence indicates that neither integration nor replication occurs.^[19,40,41] The administered plasmid DNA apparently resides transiently in the nucleus of the transfected cell as an extra-chromosomal element, where it can transcribe and translate the gene it carries. Plasmid DNA has been detected for several months after inoculation into the liver or muscle. However, there was no evidence for replication or integration into the chromosome.^[37,42,43] This was supported by the DNA of bacterial origin retaining the methylation patterns of bacteria, rather than of the mammalian host in which it was being maintained.

The stability and persistence of transferred plasmids is an important issue that is not resolved. The plasmid DNA remains unintegrated in transfected cells for the life of the cell. For dermal inoculation, the cells are shed within days, while intramuscular inoculation of the myocytes are long lasting. Plasmid DNA containing the nucleoprotein gene of influenza A virus (VIJNeo-NP) has been reported to persist for at least 18 weeks in mouse muscle, with a decrease over time.^[41]

Preliminary pharmacokinetic studies with DNA administered by different routes and methods indicate rapid elimination.^[43-45] Thus, DNA encoding asialo-orosomucoid-poly(lysine) administered to rats and mice was cleared with first order kinetics with a half life of 2.5 minutes.^[45] The intracellular events affecting gene uptake and expression have been computer modelled by Ledley and Ledley.^[46] The various parameters include: (i) distribution and biological fate of the DNA expression vector; (ii) rates of DNA uptake into cells and dynamics

of intracellular movement of the DNA to the nucleus; (iii) half-life of the DNA vector in the cell; (iv) transcription rate; (v) half-life of mRNA; (vi) translation rate; and (vii) post-translational processing, distribution and fate of the gene product. At this time, there is very little knowledge about each of these steps, especially as they apply to expression in relation to the induction of immune responses. Further studies are needed to understand the kinetics of gene uptake, expression and persistence in the host.

4. Plasmids for DNA Immunisation

The aim of the immunologist is to express high levels of a gene product defining antigenic domains in a particular mammalian cell that will stimulate the desired immune response, and to be able to modulate this response when necessary. The gene product should preferably not be toxic to these cells, as has been found with some rabies and herpes virus proteins.^[47,48] The low efficiency of direct gene transfer with plasmid DNA, when only 1 to 2% of mouse muscle cells are transfected,^[29] makes high level expression necessary. The other desirable properties of expression vectors for DNA immunisation include: (i) the plasmids should exist as a high copy number in *Escherichia coli*; and (ii) no replication or integration into the host chromosome should occur. The plasmid DNA for vaccination must be free of any contamination, particularly toxic or inherently antigenic substances including endotoxins and antibiotics such as ampicillin. Commercially available plasmids and purification methods are available to meet these requirements.^[49]

A range of viral enhancers have been successfully used in vectors for direct DNA immunisation, including those of SV40, retrovirus, polyoma and papilloma, as well as a variety of cellular enhancers. The highest levels of expression in a wide range of mammalian cells and host animals has been obtained with the enhancer from human cytomegalovirus (CMV). The enhancer, promoter and intron A from the human CMV 1E1 gene gave the highest expression of luciferase in mouse mus-

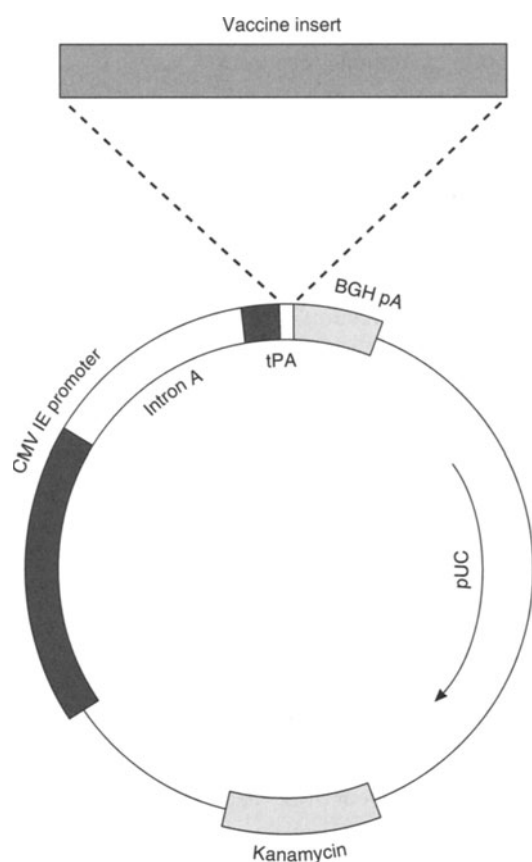


Fig. 4. Typical plasmid used for DNA vaccination. The double-stranded plasmid DNA derived from bacteria contains a strong eucaryotic promoter from human cytomegalovirus [CMV immediate-early (IE) promoter], intron A, sequences encoding a tissue plasminogen activator leader (tPA), a cloning site for inserting the gene of interest, the bovine growth hormone polyadenylation sequence (BGH pA) and a selectable marker (kanamycin).

cle^[40] and in rat tissues after gene-gun delivery.^[50] The importance of intron A from human CMV in high level expression of glycoproteins was demonstrated by Chapman et al.,^[51] who showed that the sequences in intron A positively regulate expression from the immediate-early enhancer/promoter. They also conclude that no host/vector combination is likely to be universal for good glycoprotein expression.

Another factor affecting the ability to transfect cells *in vivo* is the physical state of the DNA; cir-

cular DNA was 50- to 100-fold more efficient than linearised DNA expressing luciferase in rat heart muscle.^[52] In addition to expression of large amounts of protein for DNA immunisation, the plasmid may also require short immunostimulatory sequences.^[53] These investigators divide the process into 2 parts: a transcriptional unit that directs antigen synthesis, and an adjuvant unit in the plasmid backbone that elicits the production of type 1 interferon and IL-12 in the transfected cells and APCs. Both the location and precise sequence of the immunostimulating DNA sequences within the plasmid appear to be important for DNA immunisation. Although the short immunostimulating DNA sequences stimulated CTLs and T_H1 immune responses, they may interfere with gene expression by stimulating interferon- α (IFN α) production. Thus, these immunostimulatory sequences may be beneficial for DNA immunisation, but would not be beneficial for most gene therapy situations where an immune response is definitely undesirable. Studies using the mammalian MHC class I promoter/enhancer indicate that this system may be better than SV40 or CMV promoters for long term intracellular expression, because they are not downregulated by interferon- γ (IFN γ).^[54] On the other hand, the caveat of mammalian homology applies.

Plasmids designed for expression *in vivo* should not replicate, and should minimise the possibility for integration.^[55] To minimise the latter possibility, the transcriptional central units of the plasmids should not have homology with host cellular sequences. Thus, the use of eukaryotic cellular enhancers should be avoided for use in long-lived animals. The ultimate vector for use in DNA immunisation in humans and other animals, that will meet all of the above requirements, is clearly desirable, but has not yet been perfected. Plasmids for use in DNA immunisation will continue to be refined in the coming years. It may not be optimal for some antigens to be expressed for extended periods, and plasmids designed to express for a defined period before they self destruct may be desirable.

5. Immunity to Infectious Agents Induced by DNA Immunisation

After the initial reports of the induction of protective immunity to influenza by vaccination of chickens or mice with DNA coding for the haemagglutinin or nucleoprotein genes of influenza virus,^[10,16,55,56] there was the demonstration of the applicability of this concept to a number of other virus systems, including herpesviruses in mice and cattle,^[11,57] rabies in mice,^[58] hepatitis B surface antigen,^[12] HIV^[13,59] and lymphocytic choriomeningitis virus (LCMV)^[60-62] (table I). Subsequent studies have established that DNA immunisation is applicable to all biological systems extending from parasites^[63,64] to tumour antigens,^[65] but not directly to carbohydrate antigens. The potential benefits of DNA immunisation are:

- DNA is easy and inexpensive to prepare in a pure form, eliminating the possibility of contamination with adventitious agents from tissue culture.
- The plasmid is non-infectious as compared with replicating virus vector systems.
- No detectable immune responses are induced by the plasmid.
- It may be possible to induce immune responses in the presence of antibody, which is a problem with, for example, immunisation against measles in children who have maternal antibody.
- Immunisation is possible against agents that cannot be grown in culture.
- The vaccine is stable, making it suitable for use in developing countries where a cold chain is not available to maintain the viability of some conventional live-attenuated vaccines.

6. Advantages and General Features of DNA Immunisation

A review of the early literature (table I) reveals that DNA immunisation behaves like infection at initiating both antibodies and cell-mediated immune responses. We can then ask if there are any particular advantages to the use of DNA immunisation over conventional live attenuated vaccines,

and what this technology can teach us about the induction of immune responses.

6.1 Induction of Protective Immunity to Influenza in Mice and Chickens

In addition to demonstrating the efficacy of DNA immunisation, the initial studies with influenza in mice and chickens established that the efficiency of transfection does not necessarily determine the efficacy of vaccination.^[10,16,18] Thus, intramuscular injection is the site for optimal expression of saline-dissolved DNA *in vivo*, whereas intradermal administration was shown to be just as efficacious for inducing immunity. These studies indicate that antigen presentation is probably more important than high levels of expression. The high efficiency of gene-gun delivered DNA to the epidermis of mice (see section 2) has been hypothesised to be due to the presence of Langerhans cells that are capable of presenting transfected antigens to the T helper compartment of the immune system. These studies also established the induction of both T helper and B cell immunity, and set the stage for further development in DNA immunisation.

6.2 Induction of Crossreactive (Heterotypic) Immunity to Influenza Viruses

Influenza viruses are negatively stranded RNA viruses that have a high error rate during transcription by the polymerase genes, and have no repair mechanisms. Consequently, they show extensive genetic variation, which results in antigenic drift. This occurs mainly in the surface haemagglutinin and neuraminidase proteins.^[66] Since the haemagglutinin and neuraminidase of influenza are important for providing protective immunity, it is necessary to change human vaccine strains almost every year to keep abreast of antigenic drift. The nucleoprotein of influenza virus, one of the genetically conserved internal proteins, is known to be important in the induction of crossreactive CTLs,^[67] and is important in recovery from infection, but does not provide significant protection from initial infection. If it did, we would not have

the emergence of pandemic or even epidemic influenza viruses.

The demonstration that DNA immunisation with the nucleoprotein of influenza A provides heterotypic immunity between different subtypes of influenza A in mice^[10,55] is enormously important if this phenomenon is shown to extend to animals naturally infected with influenza, including humans, pigs, horses and chickens. Studies on the preclinical efficacy of contemporary human vaccine strains in nonhuman primates and ferrets indicates that DNA expression vectors encoding haemagglutinin and internal proteins were more efficient than conventional inactivated or subvirion vaccines.^[68] Further experiments are needed to clearly demonstrate heterotypic immunity, but if the promise is fulfilled, DNA immunisation may revolutionise the control of influenza. Preliminary studies with DNA vaccine to the nucleoprotein of A/PR/8/34 (H1N1) in mice failed to provide protection from homologous challenge,^[69] indicating that more studies are needed to determine if heterotypic immunity induced by nucleoprotein to influenza applies across all subtypes and in naturally infected hosts.

6.3 Induction of Long Term B Cell Immunity to Influenza in Mice

Comparative studies on conventional and DNA vaccines for influenza virus in mice have established that DNA vaccination is equivalent to live vaccination in protecting mice from challenge.^[70,71] Two doses of DNA vaccine administered intradermally by gene-gun provided 100% protection, as did live vaccine, while subunit vaccine provided 70% protection. The antibody-producing cells were localised in the spleen and bone marrow after vaccination, and after challenge were found in the lymph nodes draining the lungs. CD4⁺ cells were vital for induction of antibody-forming cells (AFCs); depletion of these cells with anti-CD4⁺ antibody completely abolished AFCs from all compartments. The immunity induced by DNA immunisation was maintained in the spleen and bone marrow for at least 1 year. Repeated

challenge with live influenza virus resulted in mobilisation of AFCs to the lymphoid tissues associated with the respiratory tract. Thus, DNA immunisation of mice provides long term immunity to the homologous virus, which may last a lifetime.^[72] Studies in ferrets and nonhuman primates^[68] confirm that DNA immunisation is more efficacious than conventional influenza virus subunit vaccines.

6.4 Broadening of the Antibody Response to the Haemagglutinin of Influenza Virus

There is preliminary evidence that DNA immunisation may induce more crossreactive antibody than vaccination with live influenza virus. DNA immunisation of ferrets encoding haemagglutinin of A/PR/8/34 (H1N1) induced antibodies that crossreacted with A/swine/Shope/30 (H1N1).^[73] This crossreaction was not found after natural infection of ferrets. These preliminary results warrant additional studies, because if crossreactive antibodies are induced it may not be necessary to change influenza vaccine strains so frequently. These studies indicate the potential for inducing high affinity antibodies.

6.5 Modulation of the Antibody Isotype by DNA Immunisation

Since the bacterial DNA used for DNA immunisation induces cytokines, the plasmid vector is capable of altering the intensity and character of the immune response to its encoded protein. We will first consider the character of the response. IFN γ is a central mediator of cellular immune reactions, and can promote responsiveness by up-regulating class II molecule expression. Furthermore, IFN γ participates in the interplay of T_H1/T_H2 cells and can shift the balance away from a T_H2 response. At present, the full range of cytokines induced by bacterial DNA is not known, and it is unclear whether the action of this molecule is to promote responsiveness in general or steer the immune system in the direction of T_H1 responsiveness.^[74]

DNA vaccination of mice with the haemagglutinin of influenza virus by the intramuscular and intradermal routes induced primarily IgG2a antibodies, whereas gene-gun immunisation produces mainly IgG1 antibodies. Thus, different methods of delivering gene vaccines can influence the isotype of the antibodies produced.^[75] It has been tentatively postulated that intramuscular immunisation can result in T_H1-like responses due to the predominance of IgG2a antibodies, CTL activity, IFN γ production and lack of interleukin-4 release. Gene-gun immunisation, on the other hand, appears to elicit a T_H2 response,^[76] which may be due to the low dose of DNA rather than to the site of inoculation. With high doses of DNA, as with intramuscular or intradermal injection, the CpG adjuvant contribution would be great enough to drive the response strongly towards T_H1, whereas with gene-gun there would be so little CpG that the antigen itself would largely determine whether the response was T_H1 or T_H2. Studies on antibody isotypes induced in mice after DNA immunisation with plasmid DNA encoding the circumsporozoite protein of the malarial parasite *Plasmodium yoelii* have revealed that there was an isotype switch from IgG1 to IgG2a.^[64] Isotype switching has important implications for therapeutic manipulation of the immune system, particularly as it relates to allergies and autoimmunity.

6.6 Manipulation of the Immune Response with Cytokine Genes

In addition to altering the character of the immune response, the cytokines induced by the plasmids used for DNA immunisation can alter the intensity of the response.^[74] Cytokines play a central role in the immune response by promoting the activation of specific and non-specific effector mechanisms.^[77] Granulocyte-macrophage colony stimulating factor (GM-CSF), known to facilitate induction of the primary immune response by activating and recruiting APCs,^[78,79] was tested for its effect on DNA immunisation using a plasmid expressing the glycoprotein of rabies virus.^[80] These investigators also tested the affect of mouse IFN γ

that is known to upregulate the expression of the MHC.^[81,82] Co-inoculation of mice with vectors expressing GM-CSF and rabies glycoprotein enhanced the B and T helper cell activity to rabies virus, while co-inoculation with a plasmid expressing IFN γ resulted in a decrease of the immune response to the viral antigen.

Co-expression of other cytokines, including IL-6,^[83,84] have also been shown to augment the immune response to gene vaccines encoding the haemagglutinin of influenza viruses. Thus, inoculation of mixtures of plasmids coding for antigen and cytokines, or polycistronic vectors expressing both, has the capacity to improve the efficacy of DNA immunisation. Most importantly, DNA immunisation has the potential for gaining further insight into the interplay of antigen and cytokine in modulation of immune responses.

6.7 DNA Immunisation Can Overcome Haplotype Restriction

An important problem sometimes encountered with immunisation is the failure in certain individuals to induce an adequate immune response. This is particularly troublesome for immunisation of adults against hepatitis B virus (HBV), where it is estimated that up to 25% of individuals fail to respond to the currently used recombinant protein vaccines. Studies in mice with a DNA vaccine to hepatitis B surface antigen (HBsAg) showed a strong sustained humoral and CTL response after a single injection that was superior to that obtained with recombinant protein. Subsequent studies in congenic mice revealed that DNA-based immunisation can overcome the haplotype-restricted response to HBsAg.^[85] Thus, DNA immunisation may be useful for vaccination of humans who fail to respond to recombinant HBsAg. The high levels of CTLs induced also raises the possibility that DNA immunisation may offer a therapeutic approach to chronic hepatitis B infection, for it has been shown that it is possible to downregulate HBV gene expression in transgenic mouse lines with CTL cell lines.^[86]

7. DNA Immunisation to HIV

The induction of highly efficacious humoral and cell-mediated immune responses to HBV and influenza in mice and primates shows promise for the control of a highly variable virus (influenza) and raises the possibility of control of HIV by DNA immunisation. HIV has proved to be difficult to vaccinate against because of antigen diversity and the inability of postinfection immune responses to protect against the slow, but inexorable, development of AIDS.^[87] Thus, a vaccine would be required to maintain high levels of continuously active immune responses capable of controlling a diversity of HIV-1 isolates.

Considerable work has been done using DNA to raise immune responses to immunodeficiency viruses. In murine models, plasmids expressing the HIV-1 envelope (Env) glycoprotein have raised both neutralising antibody and CTL responses.^[13,59,76] DNA inoculations in macaques have also raised antibody to the HIV-1 Env glycoprotein.^[13] The virulent, uncloned Simian immunodeficiency virus (SIV) SIV_{mac251}/Rhesus macaque model was used to evaluate the ability of DNA-expressed nonsecreted forms of the Env glycoprotein of 3 different SIV_{mac} tropic variants to protect rhesus macaques against challenge.^[88] Multiple inoculations of plasmid DNA by different routes and methods [intravenous, intramuscular, intradermal (gene-gun)] induced high levels of neutralising antibodies, but the titres were transient and were not boosted by additional plasmid DNA injections. CTL activity for Env was also raised in all of the vaccinated animals. The temporal appearance of CTLs was similar to that of antibody. However, while antibody responses fell with time, CTL responses persisted. SIV_{mac} challenge was administered intravenously at 2 weeks following the last immunisation. The DNA immunisations did not prevent infection or protect against CD4+ cell loss. Long term levels of infection were similar in the vaccinated and control animals, with 1 in 10⁴ to 10⁵ of peripheral blood cells carrying infectious virus. However, viral loads were reduced to the long term level over a shorter period of time in the vaccinated

groups (6 weeks) than in the control group (12 weeks). Thus, the DNA vaccine raised both neutralising antibody and CTL responses and provided some attenuation of the acute phase of infection, but did not prevent the loss of CD4+ cells.

The transience of anti-Env antibody responses in macaques is in keeping with an earlier HIV-1 DNA vaccine trial in mice.^[59] Transient antibody responses have also been seen in mice immunised with DNA expressing the circumsporozoite protein of malaria.^[64] These transient humoral responses are in contrast with the persistent antibody responses seen in mice immunised with DNAs expressing the influenza virus haemagglutinin and nucleoprotein,^[24,89] the HBV surface antigen,^[90] the rabies virus glycoprotein G^[58] and HIV-1.^[59] DNA-raised humoral responses to influenza virus haemagglutinin and nucleoprotein proteins also persist in African green monkeys.^[68] Although DNA immunisation did not achieve the immune responses induced by live attenuated infection with SIV_{mac}, it did show evidence of induction of broad neutralising antibody responses, but also induced enhancing antibodies. Thus, DNA immunisation partially mimicked natural infection, and additional studies to dissect the immune response to HIV gene immunisation are clearly warranted.

8. Variability in the Immune Responses Induced by DNA Immunisation

DNA immunisation varies within and between systems in the efficacy of induction of immunity (table I). It is highly efficacious at inducing immune responses to HBV and influenza in mice, but is variable at inducing immunity to influenza in chickens.^[16,18] Protection induced in mice to LCMV by DNA immunisation was less effective than when a recombinant vaccinia construct was used for expression,^[61] and DNA immunisation of mice with a herpes glycoprotein gene failed to induce a CD8+ response.^[57] It is apparent from the studies on SIV in primate models described in section 7 that DNA immunisation does not fully mimic live attenuated vaccines at eliciting protective immunity.^[88]

The variable results reported by Robinson et al.^[16] in chickens with influenza virus could reflect several factors that influence the efficiency of expression of antigen genes and the immunogenicity of DNA vaccines, including the reproducibility of inoculation and the choice of the promoter used to drive antigen gene expression. Depending on their origin, promoters differ in tissue specificity and efficiency in initiating mRNA synthesis.^[51,58] To date, most DNA vaccines in mammalian systems have relied upon viral promoters derived from CMV. These have had good efficiency in both muscle and skin inoculation in a number of mammalian species, but may not be as effective in chickens. Another factor known to affect the immune response elicited by DNA immunisation is the method of DNA delivery; parenteral routes can yield low rates of gene transfer and produce considerable variability of gene expression.^[55] High-velocity inoculation of plasmids into the epidermis, using a gene-gun, enhanced the immune responses of mice,^[18] presumably because of a greater efficiency of DNA transfection and more effective antigen presentation by dendritic cells. Finally, the variability among chicken experiments might well reflect the use of outbred birds.

We have investigated these possibilities with DNA encoding the H5 haemagglutinin of A/turkey/Ireland/83 (H5N8) with a CMV immediate-early or a chicken β -actin promoter delivered by gene-gun to the dermis of outbred chickens.^[91] The dose response curves indicated no substantial difference between p β -actin/H5 and pCMVH5 in the levels of protection induced, and there was excellent correlation between the dose of DNA administered and protection. The absence of variability in these experiments was in marked contrast to those described earlier.^[16] We postulate that the combination of the new plasmid constructs, plus the use of a more efficient delivery mechanism such as the gene-gun, was responsible for the decreased variability. The DNA-based vaccines were superior to inactivated vaccines and provided significant cross-protection among lethal avian influenza viruses within the H5 subtype.^[91] Thus, the variability

reported in some of the early experiments is probably due to technical aspects of plasmid construction or delivery, and indicates a universality to the use of DNA-based immunisation in different species. The variability simply reflects our lack of knowledge about the technical aspects of DNA immunisation and defines the challenge before us. The ability to manipulate the immune response by DNA immunisation holds exciting possibilities for 'difficult and variable antigens' like HIV, and as we dissect the immune response, the complete picture will be revealed.

9. Safety Considerations

The requirements of all vaccines are that they are: (i) safe; (ii) effective; (iii) cheap; (iv) heat stable; (v) able to induce long lasting immunity; and (vi) simple to administer.^[92] At the first symposium on nucleic acid vaccines organised by the World Health Organization, DNA immunisation was described as 'arguably the greatest advantage over the use of purified recombinant-derived protein antigens, especially in the many cases in which it has proved difficult or impossible to produce a complex viral or parasitic antigen in a native form using traditional recombinant techniques *in vitro*'.^[93] The introduction of genetic information into mammalian hosts also raises theoretical concerns about the first requirement of a vaccine, namely safety. The 3 major issues concerning the use of DNA vaccines are: (i) the formation of anti-DNA antibodies; (ii) unexpected and untoward consequences of persistent expression of a foreign antigen; and (iii) the potential for transformation events.^[94]

9.1 Anti-DNA Antibodies

Most of the information about anti-DNA antibodies stems from studies on autoimmune diseases such as systemic lupus erythematosus (SLE). Antibodies against single-stranded (ss) DNA, double-stranded (ds) RNA, polyribonucleotides, ribonucleoprotein (RNP) and ribosomes are found at high levels in the sera of patients with SLE.^[95] Autoantibodies could arise as a consequence of polyclonal B cell expansion and/or antigen-driven

B cell activation and selection, but it is probable that neither is sufficient by itself for the development of autoimmunity. In addition, whilst antibodies to ssDNA can be induced in experimental mice, it is exceedingly difficult to raise antibodies to native dsDNA. When this is achieved, it is usually denatured ssDNA, often coupled to a carrier, and not to the original native DNA. Studies to date indicate that there is no evidence for antibody induction against the DNA after repeated injections^[43,96] of mice or humans.^[65] To provide rigorous testing of the immunogenicity of plasmid DNA, autoimmune prone mice (B/W) and normal mice (Balb/c) were immunised with a malaria plasmid (*Plasmodium yoelii*).^[64] The pattern of the immune response in each type of mouse was similar, and there was no evidence that the B/W mice produced anti-DNA antibodies that interfered with vaccine effectiveness. Thus, the formation of anti-DNA antibodies and the induction of autoimmune disease were considered unlikely consequences of the administration of plasmid DNA.

9.2 Altered Immune State

These possibilities include the induction of tolerance, autoimmunity, anaphylaxis, hyperimmunity or autoaggression. At this time each of these are theoretical considerations, and so far there is no evidence supporting these possible scenarios.^[94]

9.3 Transformation

Theoretically, the introduction of extraneous DNA could cause a transformational event leading to the formation of tumour cells, by the insertion of an active oncogene, by insertional activation of a host-cell proto-oncogene or by insertional deactivation of a suppressor gene.^[97] Insertion itself can occur in 1 of 3 ways: by random integration, by homologous recombination or by retroviral insertion. The most likely means in the present context is by random integration. It is known that DNA transfected by a variety of means into actively growing cell cultures can be integrated either randomly or by homologous recombination,^[98] with random insertion being the more frequent event,

even when good homology exists. Whether similar integration occurs in nondividing muscle cells after injection of DNA vaccine plasmids, or in cells from other tissues by incidental exposure, has not been definitely determined. For DNA vaccines, insertion should specifically be avoided; the absence in the DNA construct of sequences known to promote insertion into the human chromosome would help minimise potential insertion.

The available information considered in section 3 indicates that neither integration nor replication of the injected plasmid DNA has been detected. In analysis of integration of plasmid DNA injected into muscle, DNA re-extracted from the muscle was re-cloned into *E. coli*.^[99] The study allowed identification of junctions between chromosomal and plasmid DNA that would indicate integration. In an examination of over 1800 re-cloned plasmids, no evidence for integration was detected.

To detect the presence of vaccine plasmid (VIJneo-NP) in purified host genomic DNA, the polymerase chain reaction was utilised by Nichols et al.^[41] in studies in mice. These studies demonstrated that no integration could be detected to a sensitivity of 1 to 7.5 plasmid copies per 150 000 nuclei at 3, 6, 12 and 18 weeks after injecting 100 µg of vaccine into each quadriceps muscle and 12 other tissues of mice. This level of detection indicates that any mutation frequency induced would be approximately 3 orders of magnitude less than the spontaneous mutation frequency.

A worst-case scenario of risk assessment has also been determined by Temin,^[100] using data from cell culture transfection and infection with retroviruses. Temin calculated the probability that injected DNA would have a harmful effect to be less than 10^{-16} to 10^{-19} per DNA molecule. In the case of the DNA vaccine exploratory study,^[41] an injection of 200 µg of a 6 kb plasmid is equivalent to 3×10^{13} molecules of plasmid. This represents a 10^3 to 10^6 safety margin compared with Temin's worst-case scenario.

While it is known that direct inoculation of oncogene DNA can induce tumours, the risk of induction of tumours by plasmids specifically designed

for DNA immunisation is several orders of magnitude below the natural background mutation rate for human DNA. While it is apparent that additional testing for integration will be needed before DNA vaccines can be accepted for general use in humans, the risk : benefit ratios will have to be considered as they relate to specific diseases.

9.4 Regulatory Considerations

Consideration of the factors affecting the development and certification of DNA vaccines has been dealt with in an excellent article by Hilleman.^[101] A major consideration is close cooperation between the interested parties and the regulatory control authorities who have the authority for approving the content, quality, and safety of the product and clinical testing of candidate vaccine. Previous experience with live virus vaccines may provide information on safety and benefits that are relevant to DNA vaccines. As Hilleman points out, 'it is perhaps a great benefit to mankind that the safety of live viral vaccines was established during an era of lesser technical and regulatory complexities'. It is most encouraging that DNA-based vaccines for influenza and HIV are in clinical trials in humans.^[102]

10. Mechanisms of Induction of Immunity by DNA Immunisation

DNA immunisation induces effective specific immunity to viral and parasitic diseases involving both cell-mediated T lymphocytes and antibody-mediated immunity (see section 5 and below). The available information supports the notion that DNA immunisation results in presentation of antigens to the host's immune system in a natural form, similar to that achieved with live-attenuated vaccines.

One of the reasons live-attenuated vaccines are so efficacious is that they activate the CD8+ (and CD4+) precursor cells that give rise to the major antiviral effector cells: the MHC class I restricted CTLs and the class II restricted T helper cells.^[103] The reason for this resides in the cell biology of the synthesis, assembly and transport of the MHC class

I molecule. The most effective way for this to occur is for the viral peptides to be expressed *de novo* in the cells. Direct plasmid DNA insertion into cells provides the opportunity for this to occur without viral infection.

One model for priming of humoral and cellular immune responses by DNA vaccines has reflected the distinct pathways of MHC class I and II antigen processing in cell culture.^[22] Immune responses have been proposed to occur when antigen encoded by the myocyte or keratinocyte is released by secretion or cell death. The antigens are then taken up by macrophages and B cells, thereby initiating a T helper-dependent antibody response.

A second alternative is that CTL priming occurs by endogenous proteasome-dependent processing and presentation of antigens within the transfected keratinocyte/myocyte, followed by transporter associated with antigen processing (TAP)-dependent presentation on the myocyte MHC class I molecule (fig. 5). The difficulty with this alternative, at least after intramuscular injection, is that myocytes do not possess these co-stimulatory molecules. The available evidence suggests that immune tolerance can occur in the absence of appropriate co-stimulatory signals, which may explain the higher efficiency of gene-gun immunisation. Thus, this explanation is probably an oversimplification and neglects the importance of co-stimulatory molecules and professional APCs.

As discussed above, it has been documented that treatment with drugs causes muscle degradation, followed by regeneration and increased uptake and expression of DNA into the muscle cells. It has been hypothesised that regenerating muscle, at least in the early stage, may contain more mononucleated cells, some of which can act as efficient APCs (macrophages or dendritic cells).^[12] Additionally, the level of expression of MHC class I molecules is greater on the surface of immature than of mature muscle fibres.^[104] It is known that myoblasts and myocytes express only low levels of MHC class I determinants and fail to express MHC class II molecules.^[104] Therefore, MHC class II-restricted CD4+ T cell priming may occur in the

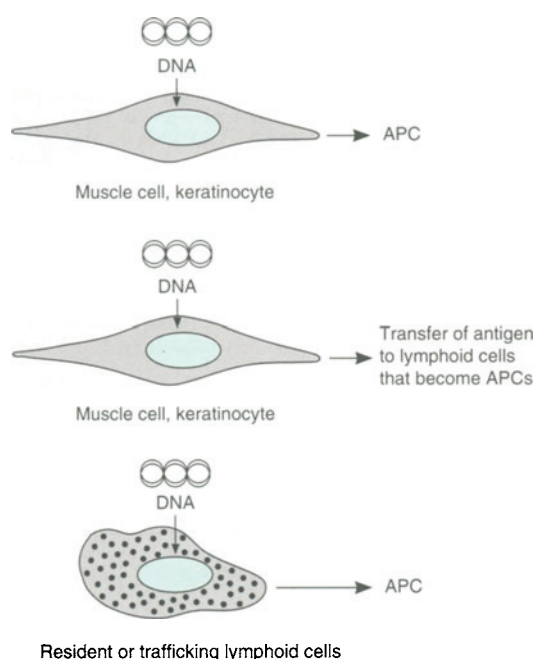


Fig. 5. Three possible models for priming of major histocompatibility complex (MHC) class I-restricted cytotoxic T lymphocyte responses by DNA injections. **Top:** direct presentation by transduced keratinocytes or muscle cells. **Middle:** transfer of protein antigens to bone marrow-derived antigen-presenting cells (APCs) with crossover into the MHC class I processing pathway. **Bottom:** transfection of DNA directly into bone marrow-derived APCs infiltrating the inoculation site, followed by classical MHC class I processing (after Pardoll & Beckerleg, [22] with permission).

APCs that infiltrate the muscle in response to injury or injection, and would have the opportunity to pick up antigen synthesised locally. The antigens could then be carried to draining lymph nodes where they could be presented to both B and T cells. These infiltrating APCs are also likely to be involved in priming of MHC class I-restricted CD8⁺ T cells. There is considerable similarity between myocyte-synthesised antigen and the presentation of tumour-specific antigens to MHC class I-restricted CD8⁺ T cells.^[105] There is evidence for the transfer of antigen to a bone marrow derived cell before its presentation to CD8⁺ T cells.^[106]

The importance of the APC in DNA immunisation was clearly demonstrated in the studies on potentiation of the response by GM-CSF.^[80] Since the mechanism of action of this cytokine is the induction of differentiation of haemopoietic progenitors of professional APCs, it demonstrates the importance of these cells in DNA immunisation. An alternative model for immunological priming in DNA immunisation is that small numbers of infiltrating macrophages (APCs) are themselves directly transfected (fig. 5).

Studies by Rouse and colleagues^[107] with DNA encoding HSV proteins support the importance of APCs in the induction of immunity by purified DNA. Using plasmid DNA encoding HSV proteins, these investigators demonstrated that when purified macrophages and dendritic cells are exposed to naked DNA *in vitro*, only the macrophages expressed proteins, as measured by Western blotting, and were able to act as APCs in primary CTL responses.^[107] However, if the same DNA is transfected into cells using the cationic lipid DOTAP, then both macrophages and dendritic cells express proteins and act as APCs. Under these conditions, dendritic cells are superior to macrophages at inducing primary CTL responses in naive T cell cultures.

The mechanism of induction of immune responses in naive animals after DNA transfection is still largely unresolved. Intradermal inoculation, either by gene-gun or by injection, has been shown to elicit higher immune responses (see section 2). As discussed above, this may result from direct inoculation of APCs and the site of immune stimulation. There is clearly much to be learned about DNA immunisation and the mechanism(s) involved. The ability to upregulate or downregulate the immune response as required lies in being able to manipulate the immune response with specifically designed plasmids. The inclusion of sequences that are mitogenic and provide adjuvant effects may stimulate desired responses, while the elimination of sequences that promote autoimmune responses or undesired cellular responses may be equally possible.

Acknowledgements

This work was supported by Public Health Service grants AI-08831 and AI-34946 from the National Institute of Allergy and Infectious Diseases, Cancer Center Support (CORE) grant CA-21765, and the American Lebanese Syrian Associated Charities (ALSAC) to RGW and Public Health Service grants AI-34946, AI-34241, AI-35149, and the US Department of Agriculture grant MASR-9303331 to HLR.

References

- Fauci AS. In: Report of the task force on microbiology and infectious diseases. Bethesda: U.S. Department of Health and Human Services, Public Health Service, National Institutes of Health, 1992: 1-55
- Gierer A, Schramm G. The infectivity of ribonucleic acid from tobacco mosaic virus. *Z Naturforsch* 1956; 116: 138-42
- Alexander HE, Koch G, Mountain JM, et al. Infectivity of ribonucleic acid from poliovirus in cell monolayers. *J Exp Med* 1958; 493:506
- Racaniello VR, Baltimore D. Cloned poliovirus complementary DNA is infectious in mammalian cells. *Science* 1981; 214: 916-9
- Fung Y-KT, Crittenden LB, Fadly AM, et al. Tumor induction by direct injection of cloned *v-src* DNA into chickens. *Proc Natl Acad Sci USA* 1983; 80: 353-7
- Benvenisty N, Reshef L. Direct introduction of genes into rats and expression of the genes. *Proc Natl Acad Sci USA* 1986; 83: 9551-5
- Atanasiu P. Production de tumeurs chez le hamster par inoculation d'acide desoxyribonucleique extrait de cultures de tissus par le virus de polyome. *CR Hebd Seances Acad Sci Ser D* 1962; 254: 4228-30
- Israel MA, Chan HW, Hourihan SL, et al. Biological activity of polyoma viral DNA in mice and hamsters. *J Virol* 1979; 29: 990-6
- Wolff JA, Malone RW, Williams P, et al. Direct gene transfer into mouse muscle *in vivo*. *Science* 1990; 247: 1465-8
- Ulmer JB, Donnelly JJ, Parker SE, et al. Heterologous protection against influenza by injection of DNA encoding a viral protein. *Science* 1993; 259: 1745-9
- Cox G, Zamb TJ, Babiuk LA. Bovine herpesvirus 1: immune responses in mice and cattle injected with plasmid DNA. *J Virol* 1993; 67: 5664-7
- Davis HL, Michel M, Whalen R. DNA-based immunization induces continuous secretion of hepatitis B surface antigen and high levels of circulating antibody. *Hum Mol Genet* 1993; 2: 1847-51
- Wang B, Ugen KE, Rikantan V, et al. Gene inoculation generates immune responses against immunodeficiency virus type 1. *Proc Natl Acad Sci USA* 1993; 90: 4156-60
- Tang D-C, DeVit M, Johnston SA. Genetic immunization is a simple method for eliciting an immune response. *Nature* 1992; 356: 152
- Hunt LA, Brown DW, Robinson HL, et al. Retrovirus-expressed hemagglutinin protects against lethal influenza virus infections. *J Virol* 1988; 62: 3014-9
- Robinson HL, Hunt LA, Webster RG. Protection against a lethal influenza virus challenge by immunization with a haemagglutinin-expressing plasmid DNA. *Vaccine* 1993; 11: 957-60
- Vahlsing HL, Yankauckas MA, Sawdey M, et al. Immunization with plasmid DNA using a pneumatic gun. *J Immunol Methods* 1994; 175: 11-22
- Fynan EF, Webster RG, Fuller DH, et al. DNA vaccines: protective immunizations by parenteral, mucosal, and gene-gun inoculations. *Proc Natl Acad Sci USA* 1993; 90: 11478-82
- Freeman GJ, Boriello F, Hodes RH, et al. Uncovering of a functional alternative CTL-4 counter-receptor in B7-deficient mice. *Science* 1993; 262: 907-8
- Hatchcock KS, Lazlo G, Dickler HB, et al. Identification of an alternative CTL-4 ligand costimulatory for T cell activation. *Science* 1993; 262: 905-7
- Linsley P, Brady W, Grosmaire L, et al. Binding of the B cell activation antigen B7 to CD28 co-stimulates T cell proliferation and interleukin 2 mRNA accumulation. *J Exp Med* 1991; 173: 721-30
- Pardoll DM, Beckerleg AM. Exposing the immunology of naked DNA vaccines. *Immunity* 1995; 3: 165-9
- Eisenbraun MD, Fuller DH, Haynes JH. Examination of parameters affecting the elicitation of humoral immune responses by particle bombardment-mediated genetic immunization. *DNA Cell Biol* 1993; 12: 791
- Raz E, Carson DA, Parker SE, et al. Intradermal gene immunizations: the possible role of DNA uptake in the induction of cellular immunity to viruses. *Proc Natl Acad Sci USA* 1994; 91: 9519-23
- Hengge UR, Chan EF, Foster RA, et al. Cytokine gene expression in epidermis with biological effects following injection of naked DNA. *Nature Genet* 1995; 10: 161
- Condon C, Watkins SC, Celluzzi CM, et al. DNA-based immunization by *in vivo* transfection of dendritic cells. *Nat Med* 1996; 2: 1122-8
- Ledley FD. Nonviral gene therapy: the promise of genes as pharmaceutical products. *Hum Gene Ther* 1995; 6: 1129-44
- Danko I, Fritz JD, Jiao S, et al. Pharmacological enhancement of *in vivo* foreign gene expression in muscle. *Gene Ther* 1994; 1: 114-21
- Davis HL, Whalen RG, Demeneix BA. Direct gene transfer into skeletal muscle *in vivo*: factors affecting efficiency of transfer and stability of expression. *Hum Gene Ther* 1993; 4: 151-9
- Yang N-S, Burkholder J, Roberts B, et al. *In vivo* and *in vitro* gene transfer to mammalian somatic cells by particle bombardment. *Proc Natl Acad Sci USA* 1990; 87: 9568-72
- Williams RS, Johnston SA, Riedy M, et al. Introduction of foreign genes into tissues of living mice by DNA-coated microprojectiles. *Proc Natl Acad Sci USA* 1991; 88: 2726-30
- Cheng L, Ziegelhoffer PR, Yang NS. *In vivo* promoter activity and transgene expression in mammalian somatic tissues evaluated by using particle bombardment. *Proc Natl Acad Sci USA* 1993; 90: 4455-9
- Pertmer TM, Eisenbraun MD, McCabe D, et al. Gene-gun-based nucleic acid immunization: elicitation of humoral and cytotoxic T lymphocyte responses following epidermal delivery of nanogram quantities of DNA. *Vaccine* 1995; 13: 1427-30
- Sizemore DR, Branstrom AA, Sadoff JC. Attenuated *Shigella* as a DNA delivery vehicle for DNA-mediated immunization. *Science* 1995; 270: 299-302
- Furth PA, Shamay A, Wall RJ, et al. Gene transfer into somatic tissues by jet injection. *Anal Biochem* 1992; 20: 365-8
- Wolff JA, Dowty ME, Jiao S, et al. Expression of naked plasmids by cultured myotubes and entry of plasmids into T tubules and caveolae of mammalian skeletal muscle. *J Cell Sci* 1992; 103: 1249-59

37. Sikes M, O'Malley Jr BW, Finegold MJ, et al. *In vivo* gene transfer into rabbit thyroid follicular cells by direct DNA injection. *Hum Gene Ther* 1994; 5: 837-44
38. Yovandich J, O'Malley Jr BW, Sikes M, et al. Gene transfer to synovial cells by intra-articular administration of plasmid DNA. *Hum Gene Ther* 1995; 6: 603-10
39. Coffin JM. Molecular mechanisms of nucleic acid integration. *J Med Virol* 1990; 31: 43-9
40. Manthorpe M, Cornefert-Jensen F, Hartikka J, et al. Gene therapy by intramuscular injection of plasmid DNA: studies on firefly luciferase gene expression in mice. *Hum Gene Ther* 1993; 4: 419-31
41. Nichols W, Ledwith B, Manan S, et al. Potential DNA vaccination into host cell genome. *Ann NY Acad Sci* 1995; 772: 30-9
42. Wilson JM, Grossman M, Cabrera JA, et al. A novel mechanism for achieving transgene persistence *in vivo* after somatic gene transfer into hepatocytes. *J Biol Chem* 1992; 267: 11483-9
43. Stankovics J, Andrews E, Crane AM, et al. Overexpression of human methylmalonyl CoA mutase in mice after *in vivo* gene transfer with asialo-glycoprotein/polylysine/DNA complexes. *Hum Gene Ther* 1994; 5: 1095-104
44. Wolff JA, Williams P, Acsadi G, et al. Conditions affecting direct gene transfer into rodent muscle *in vivo*. *Biotechnology* 1991; 11: 474-85
45. Lew D, Parker SE, Latimer T, et al. Cancer gene therapy using plasmid DNA: pharmacokinetics study of DNA following injection in mice. *Hum Gene Ther* 1995; 6: 553-64
46. Ledley TAS, Ledley FD. A multicompartment, numerical model of cellular events in the pharmacokinetics of gene therapies. *Hum Gene Ther* 1994; 5: 679-91
47. Ertl HJ, Verma P, He Z, et al. Plasmid vectors as anti-viral vaccines. *Ann NY Acad Sci* 1995; 772: 77-87
48. Friedman HM, Yee A, Diggelmann H, et al. Use of a glucocorticoid-inducible promoter for expression of herpes simplex virus type 1 glycoprotein gC1, a cytotoxic protein in mammalian cells. *Mol Cell Biol* 1989; 9: 2303-14
49. Schorr J, Moritz P, Seddon T, et al. Plasmid DNA for human gene therapy and DNA vaccines: production and quality assurance. *Ann NY Acad Sci* 1995; 772: 271-3
50. Cheng L, Ziegelhoffer PR, Yang N-S. *In vivo* promoter activity and transgene expression in mammalian somatic tissues evaluated by using particle bombardment. *Proc Natl Acad Sci USA* 1993; 90: 4455-9
51. Chapman BS, Thayer RM, Vincent KA, et al. Effect of intron A from human cytomegalovirus (Towne) immediate-early gene on heterologous expression in mammalian cells. *Nucleic Acids Res* 1991; 19: 3979-86
52. Kass-Eisler A, Li K, Leinwand LA. Prospects for gene therapy with direct injection of polynucleotides. *Ann NY Acad Sci* 1995; 772: 232-40
53. Sato Y, Roman M, Tighe H, et al. Immunostimulatory DNA sequences necessary for effective intradermal gene immunization. *Science* 1996; 273: 352-4
54. Harms JS, Splitter GA. Interferon-gamma inhibits transgene expression driven by SV40 or CMV promoters but augments expression driven by the mammalian MHC I promoter. *Hum Gene Ther* 1995; 6: 1291-7
55. Montgomery DL, Shiver JW, Leander KR, et al. Heterologous and homologous protection against influenza A by DNA vaccination: optimization of DNA vectors. *DNA Cell Biol* 1993; 12: 777-83
56. Robinson HL, Hunt LA, Webster RG. Protection against a lethal influenza virus challenge by immunization with a hemagglutinin-expressing plasmid DNA. *American Society for Virology 1992 Annual Meeting*: 1992 Jul 11-15; Ithaca (NY), 14
57. Manickan E, Rouse RJD, Yu Z, et al. Genetic immunization against herpes simplex virus: protection is mediated by CD4⁺ T lymphocytes. *J Immunol* 1995; 155: 259-65
58. Xiang ZQ, Spitalnik S, Tran M, et al. Vaccination with a plasmid vector carrying the rabies virus glycoprotein gene induces protective immunity against rabies virus. *Virology* 1994; 199: 132-40
59. Lu S, Santoro JC, Fuller DH, et al. Use of DNAs expressing HIV-1 env and noninfectious HIV-1 particles to raise antibody responses in mice. *Virology* 1995; 209: 147-54
60. Martins LP, Lau LL, Asano MS. DNA vaccination against persistent viral infection. *J Virol* 1995; 69: 2574-82
61. Yokoyama M, Zhang J, Whitton JL. DNA immunization confers protection against lethal lymphocytic choriomeningitis virus infection. *J Virol* 1995; 69: 2684-8
62. Zarozinski CC, Fynan EF, Selin LK, et al. Protective CTL-dependent immunity and enhanced immunopathology in mice immunized by particle bombardment with DNA encoding an internal virion protein. *J Immunol* 1995; 154: 4010-7
63. Sedegah M, Hedstrom R, Hobart P, et al. Protection against malaria by immunization with plasmid DNA encoding circumsporozoite protein. *Proc Natl Acad Sci USA* 1994; 91: 9866-70
64. Mor G, Klinman DM, Shapiro S, et al. Complexity of the cytokine and antibody response elicited by immunizing mice with *Plasmodium yoelii* circumsporozoite protein plasmid DNA. *J Immunol* 1995; 155: 2039-46
65. Nabel GJ, Nabel EG, Yang ZY, et al. Direct gene transfer with DNA-liposome complexes in melanoma: expression, biologic activity, and lack of toxicity in humans. *Proc Natl Acad Sci USA* 1993; 90: 11307-11
66. Murphy BR, Webster RG. Orthomyxoviruses. In: Fields BN, Knipe DM, Howley PM, et al., editors. *Fields' virology*. 3rd ed. New York: Raven Press, 1996: 1397-445
67. Andrews ME, Coupar BEH. Efficacy of influenza haemagglutinin and nucleoprotein as protective antigens against influenza virus infection in mice. *Scand J Immunol* 1988; 28: 81-5
68. Donnelly JJ, Friedman A, Martinez D, et al. Preclinical efficacy of a prototype DNA vaccine: enhanced protection against antigenic drift in influenza virus. *Nat Med* 1995; 1: 583-7
69. Robinson HL, Boyle CA, Feltquate DM, et al. DNA immunization for influenza virus: studies using hemagglutinin and nucleoprotein-expressing DNAs. *J Infect Dis* 1997. In press
70. Justewicz DM, Morin MJ, Robinson HL, et al. Antibody-forming cell response to virus challenge in mice immunized with DNA encoding the influenza virus hemagglutinin. *J Virol* 1995; 69: 7712-7
71. Boyle CM, Morin M, Webster RG, et al. Role of different lymphoid tissues in the initiation and maintenance of DNA-raised antibody responses to the influenza virus H1 glycoprotein. *J Virol* 1996; 70: 9074-8
72. Justewicz DM, Webster RG. Long-term maintenance of B cell immunity to influenza virus hemagglutinin in mice following DNA-based immunization. *Virology* 1996; 224: 10-7
73. Webster RG, Fynan EF, Santoro JC, et al. Protection of ferrets against influenza challenge with a DNA vaccine to the haemagglutinin. *Vaccine* 1994; 12: 1495-8
74. Pisetsky DS, Reich C, Crowley SD, et al. Immunological properties of bacterial DNA. *Ann NY Acad Sci* 1995; 772: 152-63

75. Feltquate DM, Heaney S, Webster RG, et al. Different TH cell-types and antibody isotypes generated by saline and gene-gun DNA immunization. *J Immunol* 1997; 158: 2278-84
76. Fuller DH, Haynes JR. A qualitative progression in HIV type 1 glycoprotein 120-specific cytotoxic cellular and humoral immune responses in mice receiving a DNA-based glycoprotein 120 vaccine. *AIDS Res Hum Retrovir* 1996; 10: 1433-41
77. Trout AB, Kelso A. Cytokine genes, regulation of. In: Roitt IM, Dives PJ, editors. *Encyclopedia of immunology*. London: Academic Press, 1992: 434-6
78. Fischer HG, Frosch S, Reske K, et al. Granulocyte-macrophage colony-stimulating factor activates macrophages derived from bone marrow cultures to synthesis of MHC class II molecules and to augmented antigen presentation function. *J Immunol* 1988; 141: 3882-8
79. Heufler C, Koch F, Schuler G. Granulocyte-macrophage colony-stimulating factor and interleukin 1 mediate the maturation of murine epidermal Langerhans cells into potent immunostimulatory cells. *J Exp Med* 1988; 167: 700-5
80. Xiang Z, Ertl HCJ. Manipulation of the immune response to a plasmid-encoded viral antigen by coinoculation with plasmids expressing cytokines. *Immunity* 1995; 2: 129-35
81. Steeg PS, Moore RN, Johnson HM. Regulation of murine macrophage Ia antigen expression by a lymphokine with immune interferon activity. *J Exp Med* 1982; 156: 1780-93
82. Fabry Z, Raine CS, Hart MN. Nervous tissue as an immune compartment: the dialect of the immune response in the CNS. *Immunol Today* 1994; 15: 218-24
83. Ramshaw IA, Leong KH, Ramsay AJ, et al. Novel vaccine strategies using recombinant fowlpox viruses expressing cytokine genes. *Molecular Approaches to the Control of Infectious Diseases: 1994 Oct 5-9*; Cold Spring Harbor (NY), 116
84. Ramshaw IA, Leong KH, Ramsay AJ, et al. Induction of protective immunity against influenza virus using DNA and recombinant avipoxvirus vectors. In: Brown LE, Hampson AW, Webster RG, editors. *Options for the control of influenza III*. Amsterdam: Elsevier Science, 1996: 772-6
85. Davis HL, Mancini M, Michel M-L, et al. DNA-based immunization overcomes H-2 haplotype restricted non-responsiveness to hepatitis B surface antigen in mice. *Molecular Approaches to the Control of Infectious Diseases: 1994 Oct 5-9*; Cold Spring Harbor (NY), 119
86. Guidotti LG, Ando K, Hobbs MV, et al. Cytotoxic T lymphocytes inhibit hepatitis B virus gene expression by a non-cytolytic mechanism in transgenic mice. *Proc Natl Acad Sci USA* 1994; 91: 3764-8
87. Bolognesi DP. The dilemma of developing and testing AIDS vaccines. In: Cooper BM, Temin RG, Suden B, editors. *The DNA provirus: Howard Temin's scientific legacy*. Washington (DC): American Society for Microbiology, 1995: 301-12
88. Lu S, Arthos J, Montefiori DC, et al. Simian immunodeficiency virus DNA vaccine trial in macaques. *J Virol* 1996; 70: 3978-91
89. Robinson HL, Feltquate DM, Morin MJ. DNA vaccines: a new approach to immunization. In: Brown F, Chanock R, Ginsberg H, editors. *Vaccines 95*. Cold Spring Harbor (NY): Cold Spring Harbor Laboratory Press, 1995: 69-75
90. Michel M-L, Davis HL, Schleef M, et al. DNA-mediated immunization to the hepatitis B surface antigen in mice: aspects of the humoral response mimic hepatitis B viral infection in humans. *Proc Natl Acad Sci USA* 1995; 92: 5307-11
91. Kodihalli S, Haynes JR, Robinson HL, et al. Cross protection among lethal H5N2 influenza viruses induced by DNA vaccine to the hemagglutinin. *J Virol* 1997; 71: 3391-6
92. Ada GL. Vaccines and immune responses. In: Webster RG, Granoff A, editors. *Encyclopedia of virology*. London: Academic Press, 1994: 1503-7
93. SchYdel F, Aguado M-T, Lambert P-H. Introduction: nucleic acid vaccines. *Vaccine* 1994; 12: 1491-2
94. Robertson JS. Safety considerations for nucleic acid vaccines. *Vaccine* 1994; 12: 1526-8
95. Koffler D, Carr RI, Agnello V, et al. Antibodies to polynucleotides: distribution in human serums. *Science* 1969; 166: 1648-9
96. Parker SE, Vahlsing LH, Serfilippi LM, et al. Cancer gene therapy using plasmid DNA: safety evaluation in rodents and non-human primates. *Hum Gene Ther* 1995; 6: 575-90
97. Griffiths E. Assuring the safety and efficacy of DNA vaccines. *Ann NY Acad Sci* 1995; 772: 152-63
98. Hasty P, Rivera-Perez J, Bradley A. The length of homology required for gene targeting in embryonic stem cell. *Mol Cell Biol* 1991; 11: 5586-91
99. Wolff JA, Ludtke JJ, Acsadi G, et al. A long-term persistence of plasmid DNA and foreign gene expression in mouse muscle. *Hum Mol Genet* 1992; 1: 363-9
100. Temin HM. Overview of biological effects of addition of DNA molecules to cells. *J Med Virol* 1990; 31: 13-7
101. Hilleman MR. DNA vectors: precedents and safety. *Ann NY Acad Sci* 1995; 772: 1-14
102. Conference Report. First humans receive naked HIV DNA vaccine. *AIDS Weekly Plus* 1996; 1: 5
103. Braciale TJ. Naked DNA and vaccine design. *Trends Microbiol* 1993; 1: 323-4
104. Hohlfeld R, Engel AG. The immunobiology of muscle. *Immunol Today* 1994; 15: 269-74
105. Bevan MJ. Cross-priming for a secondary cytotoxic response to minor H antigens with H-2 congenic cells which do not cross-react in the cytotoxic assay. *J Exp Med* 1976; 143: 1283-8
106. Kovacsics-Bankowski M, Rock KL. A phagosome-to-cytosol pathway for exogenous antigens presented on MHC class I molecules. *Science* 1995; 267: 243-6
107. Rouse BT, Nair S, Rouse RJD, et al. DNA vaccines and immunity to herpes simplex virus. *Curr Top Microbiol Immunol* 1997. In press
108. Coney L, Wang B, Ugen KE, et al. Facilitated DNA inoculation induces anti-HIV-1 immunity *in vivo*. *Vaccine* 1994; 12: 1545-50
109. Hoffman SL, Sedegah M, Hedstrom RC. Protection against malaria by immunization with a *Plasmodium yoelii* circumsporozoite protein nucleic acid vaccine. *Vaccine* 1994; 12: 1529-33
110. Lowrie DB, Tascon RE, Colston MJ, et al. Towards a DNA vaccine against tuberculosis. *Vaccine* 1994; 12: 1537-40

Correspondence and reprints: Dr Robert G. Webster, Department of Virology and Molecular Biology, St. Jude Children's Research Hospital, 332 N. Lauderdale, Memphis, TN 38105, USA.

E-mail: robert.webster@stjude.org